A scale-free stopping criterion for the stabilized porous Navier–Stokes iteration

Companion note to `article.tex` — porous Navier–Stokes VMS solver
(Casas, González-Usúa, Codina, de-Pouplana)

July 24, 2026

Abstract

This note records, in the notation of the paper `article.tex`, the stopping criterion for the outer nonlinear (Picard / Newton) iteration of the stationary stabilized algorithm (paper, `alg:StationarySystem`). The goal is a criterion *computable from the current iterate together with material and mesh data alone*, with *no* a-priori characteristic scales: no characteristic velocity U, length L, pressure P, nor any global Reynolds Re or Damköhler Da number. Those quantities are bespoke to a manufactured-solution (MMS) test where they happen to be known; a production criterion cannot assume them. We define a momentum measure ε_{M} and a mass measure ε_{C}, each a residual normalized by a quantity measured from the iterate, prove the guaranteed bound $\varepsilon_{\mathrm{C}} \leq \sqrt{d}$ for the flux-gradient form, catalogue the edge cases, and document the optional (non-default, non-implemented) pressure-normalized fallback. A closing remark records how the production code specializes these definitions.

1 Summary and the hard constraint

Stop the outer loop when

$$\text{converged} \quad \Longleftrightarrow \quad \max(\varepsilon_{\mathrm{M}}, \varepsilon_{\mathrm{C}}) \ \leq\ \texttt{tol} \qquad (\text{or with separate } \texttt{tol}_{\mathrm{M}},\ \texttt{tol}_{\mathrm{C}}), \tag{1}$$

with

- **Momentum** $\varepsilon_{\mathrm{M}} = \dfrac{\|r_{\mathrm{M}}^k\|}{D_{\mathrm{M}}^k}, \quad D_{\mathrm{M}}^k = \|\alpha\, \boldsymbol{u}\cdot\nabla \boldsymbol{u}\| + \left\| 2\,\nabla\cdot(\alpha\nu \overset{\mathrm{DS}}{\Pi} \nabla \boldsymbol{u}) \right\| + \|\alpha \nabla p\| + \|\sigma(\boldsymbol{u})\,\boldsymbol{u}\| + \|\boldsymbol{f}\|$;
- **Mass** $\varepsilon_{\mathrm{C}} = \dfrac{\|\nabla\cdot(\alpha \boldsymbol{u}^k)\|}{\|\nabla(\alpha \boldsymbol{u}^k)\|}$ with the guaranteed bound $\varepsilon_{\mathrm{C}} \leq \sqrt{d}$ (section 4; this flux-gradient form is now the *diagnostic* `eps_C_strong` — see section 8).

All norms are global $L^2(\Omega)$, $\|w\| = \big(\int_\Omega |w|^2 \,\mathrm{d}\Omega\big)^{1/2}$, accumulated by quadrature in a single assembly pass. Here $\overset{\mathrm{DS}}{\Pi}$ is the deviatoric–symmetric projector of the viscous term (paper notation) and $d \in \{2, 3\}$. Report ε_{M} and ε_{C} separately at every iteration.

Remark 1 (Hard constraint — do not violate). The criterion *must be computable from the current iterate $U^k = (\boldsymbol{u}^k, p^k)$ and known material/mesh data alone*: the porosity $\alpha(\boldsymbol{x})$, the viscosity ν, the resistance $\sigma = a(\alpha) + b(\alpha)|\boldsymbol{u}|$, the body force $\boldsymbol{f}$, and the element size h_K. *No* a-priori characteristic scale may be introduced, requested, or hard-coded — no U, L, P, and no global Re or Da. Those are bespoke to a manufactured-solution test where they happen to be known; a production criterion cannot assume them. If a scale is needed, it is *measured from the iterate*, not supplied. (Element-level Re_h, Da_h built from h_K and the iterate are allowed — they are not a-priori scales, and appear only in the optional fallback of section 6.)

2 Governing equations and residuals

Strong form (kinematic; density absorbed into ν, p, σ), following (paper, `eq:StrongMomentumEquation`–`eq:StrongMassEquation`):

$$\begin{aligned} \text{momentum:}\quad & \alpha\,\boldsymbol{u}\cdot\nabla\boldsymbol{u} - 2\,\nabla\cdot(\alpha\nu\overset{\mathrm{DS}}{\Pi}\nabla\boldsymbol{u}) + \alpha\nabla p + \sigma(\alpha,\boldsymbol{u})\,\boldsymbol{u} = \boldsymbol{f}, && (2)\\ \text{mass:}\quad & \varepsilon\, p + \nabla\cdot(\alpha\boldsymbol{u}) = 0, && (3) \end{aligned}$$

with the Darcy–Brinkman–Forchheimer closure (paper, `eq:DBFResistanceTerm`)

$$\sigma(\alpha,\boldsymbol{u}) = a(\alpha) + b(\alpha)\,|\boldsymbol{u}| \tag{4}$$

(Darcy + Forchheimer), $\varepsilon \geq 0$ a small compressibility (the iterative penalty), equal-order P_k/P_k velocity–pressure interpolation, and ASGS or OSGS VMS stabilization. We write

- **momentum residual** r_{M}^k: the momentum residual evaluated at U^k;
- **mass residual** $r_{\mathrm{C}}^k = \nabla\cdot(\alpha\boldsymbol{u}^k)$ ($+\varepsilon p^k$ if $\varepsilon > 0$; negligible — see section 5, edge case 7).

3 Momentum normalization

Numerator $\|r_{\mathrm{M}}^k\|$ — Philosophy A (algebraic, recommended, current). Measure what the solver actually drives to zero: the norm of the assembled *stabilized* nonlinear residual *vector* (ASGS/OSGS, subscales included), velocity block. Use a fixed vector norm (Euclidean or mass-matrix-weighted) — the *same* norm for every term in D_{M}. Build the denominator terms by assembling each physical term's velocity-block contribution *through the same weak form*: the viscous term is then the integrated-by-parts $\langle\nabla^{\mathrm{S}}\boldsymbol{v},\, 2\alpha\nu\overset{\mathrm{DS}}{\Pi}\nabla\boldsymbol{u}^k\rangle$, a first-derivative quantity — *no second derivatives, no special-casing by polynomial order.*

Numerator — Philosophy B (pointwise force density, only for a force-balance picture). Use $L^2(\Omega)$ norms of the strong-form force-density fields directly; r_{M}^k is then the strong residual field. Caveat: the strong residual of a finite element solution does *not* vanish at the discrete solution — it floors at the truncation level $\mathcal{O}(h^k)$ in a negative norm — so `tol` must be set above that floor. The viscous term then needs a second derivative. Prefer A.

Denominator D_{M}^k — dynamic term-magnitude envelope.

$$D_{\mathrm{M}}^k = \underbrace{\left\|\alpha\,\boldsymbol{u}^k\cdot\nabla\boldsymbol{u}^k\right\|}_{\text{convection}} + \underbrace{\left\|2\,\nabla\cdot(\alpha\nu\overset{\mathrm{DS}}{\Pi}\nabla\boldsymbol{u}^k)\right\|}_{\text{viscous}} + \underbrace{\left\|\alpha\nabla p^k\right\|}_{\text{pressure gradient}} + \underbrace{\left\|\sigma(\boldsymbol{u}^k)\,\boldsymbol{u}^k\right\|}_{\text{resistance (Darcy + Forchheimer)}} + \underbrace{\|\boldsymbol{f}\|}_{\text{body force}}. \tag{5}$$

Rationale.

- *Why a sum of term magnitudes is regime-robust.* At the solution the momentum terms do not vanish — they *balance*. Whichever mechanism dominates (viscous as $Re, Da \to 0$, convection as $Re \to \infty$, resistance as $Da \to \infty$) automatically enters the sum and sets the scale, with no regime parameter to choose. (Offline check: non-dimensionalizing sends the four operator terms to coefficients Re, 1, $(1 + Re + Da)$, Da, so $D_\mathrm{M} \sim (1 + Re + Da)\,\alpha_\infty \nu U/L^2$. The pressure-gradient term alone carries the coefficient $1 + Re + Da$ and is leading-order in *every* regime — so $\|\alpha \nabla p^k\|$ by itself is already a valid scale once p^k has developed.)

- *Why dynamic (recomputed each iteration).* σ depends on $|\boldsymbol{u}^k|$ through Forchheimer and α varies in space, so the true local resistance is *not* the nominal global Da. The dynamic sum reads the actual local magnitudes; a frozen analytic prefactor cannot.

- *Why include $\|\boldsymbol{f}\|$.* In forced / manufactured problems $\boldsymbol{f}$ is one of the balanced forces and can dominate; it prevents the scale collapsing if the operator terms cancel. When $\boldsymbol{f} = 0$ it contributes nothing.

- *Why the viscous term is unproblematic under Philosophy A.* The weak form already integrated it by parts, so it is first-order. Under Philosophy B with P_1, $\nabla\cdot(\alpha\nu \overset{\mathrm{DS}}{\Pi} \nabla \boldsymbol{u}_h) \approx 0$ element-wise; accept the ≈ 0 contribution — the co-dominant pressure-gradient term supplies the viscous scale. Do *not* patch with an inverse-estimate factor C_inv/h: it overestimates the viscous force on fine meshes and makes ε_M over-optimistic.

4 Mass normalization (flux-gradient ratio)

This section defines what is now the diagnostic `eps_C_strong`*; the production gate uses the algebraic mass residual of section 8.*

Numerator $\|\nabla\cdot(\alpha \boldsymbol{u}^k)\|$. Use the divergence of the porous flux $\boldsymbol{q}^k = \alpha \boldsymbol{u}^k$ — the quantity that $\to 0$ at the fixed point. With the iterative penalty, the εp^{n-1} term cancels at convergence, leaving the incompressibility residual. If $\varepsilon > 0$ the consistent numerator is $\|\varepsilon p^k + \nabla\cdot(\alpha \boldsymbol{u}^k)\|$, but $\varepsilon^* \sim 10^{-4}$, so $\|\nabla\cdot(\alpha \boldsymbol{u}^k)\|$ alone is equivalent and clearer.

Denominator $\|\nabla(\alpha \boldsymbol{u}^k)\|$ — the flux-gradient (Frobenius) norm.

$$\nabla(\alpha \boldsymbol{u}^k) = \alpha \nabla \boldsymbol{u}^k + \boldsymbol{u}^k \otimes \nabla\alpha \qquad \text{(full second-order tensor; Frobenius norm)}, \qquad \varepsilon_\mathrm{C} = \frac{\|\nabla\cdot(\alpha \boldsymbol{u}^k)\|}{\|\nabla(\alpha \boldsymbol{u}^k)\|} \le \sqrt{d}. \tag{6}$$

Proposition 1 (The $\sqrt{d}$ bound)**.** *Let $\boldsymbol{q} = \alpha \boldsymbol{u}^k$ and let $\nabla \boldsymbol{q}$ be its (second-order) gradient tensor. Then $\varepsilon_\mathrm{C} \le \sqrt{d}$ always.*

Proof. Pointwise, the divergence is the trace of the gradient tensor, $\nabla\cdot\boldsymbol{q} = \mathrm{tr}(\nabla \boldsymbol{q})$. For any $d \times d$ matrix A, the Cauchy–Schwarz inequality applied to $\mathrm{tr}(A) = \sum_{i=1}^d A_{ii} = \langle \mathrm{diag}, A\rangle$ gives $(\mathrm{tr}\,A)^2 \le d\,(A\!:\!A)$, where $A\!:\!A = \sum_{ij} A_{ij}^2$ is the squared Frobenius norm. Hence, pointwise,

$$|\nabla\cdot\boldsymbol{q}|^2 = \big(\mathrm{tr}(\nabla \boldsymbol{q})\big)^2 \;\le\; d\,(\nabla \boldsymbol{q} : \nabla \boldsymbol{q}) = d\,|\nabla \boldsymbol{q}|_F^2. \tag{7}$$

Integrating (7) over Ω yields $\|\nabla\cdot\boldsymbol{q}\|^2 \le d\,\|\nabla \boldsymbol{q}\|_F^2$, and dividing by $\|\nabla \boldsymbol{q}\|_F^2$ gives $\varepsilon_\mathrm{C}^2 \le d$, i.e. $\varepsilon_\mathrm{C} \le \sqrt{d}$. □

Rationale.

- *Why it is the right scale-free measure.* It is the pressure-normalized "relative pressure error from mass imbalance" with the divergence→pressure conversion factor *measured from the iterate instead of supplied.* The conversion factor $\Phi/\alpha_\infty = \|p^k\| \,/\, \|\nabla(\alpha\boldsymbol{u}^k)\|$ reproduces the full viscous+inertial+Darcy envelope in every regime; substituting it into $\varepsilon_\mathrm{C} = (\Phi/\alpha_\infty)\,\|\nabla\cdot(\alpha\boldsymbol{u}^k)\| \,/\, \|p^k\|$ cancels $\|p^k\|$ identically, leaving the flux-gradient ratio. So the gradient ratio *is* the pressure idea with the a-priori factor removed; it never requires p.

- *Why genuinely scale-free.* Both norms are built from $\boldsymbol{u}^k$ and the known field α. Dimensionless by construction.

- *Why the $\sqrt{d}$ bound (a self-check).* By proposition 1, $\varepsilon_\mathrm{C} \le \sqrt{d}$ always; a computed value above $\sqrt{d}$ signals a quadrature / projection / assembly bug — assert or log.

- *Why keep the $\boldsymbol{u}\otimes\nabla\alpha$ term.* It is the real flux structure forced by the porosity gradient; the ratio then measures dilatation against the *total* flux variation. When α is constant (including $\alpha \equiv 1$, classical Navier–Stokes) it vanishes and $\varepsilon_\mathrm{C} \to \|\nabla\cdot\boldsymbol{u}\| \,/\, \|\nabla\boldsymbol{u}\|$ — graceful, no special-casing.

- *Why global, not pointwise.* A pointwise ratio blows up wherever $\nabla(\alpha\boldsymbol{u}) \to 0$. Use domain-integrated L^2 norms. With no-slip walls plus a parabolic inlet there is always boundary-layer shear, so $\|\nabla(\alpha\boldsymbol{u}^k)\|$ is robustly bounded away from zero in practice.

5 Edge cases (apply all)

1. **Trivial initial guess $\boldsymbol{u}^0 = 0$.** D_M^0 and $\|\nabla(\alpha\boldsymbol{u}^0)\|$ may be $0 \Rightarrow 0/0$. Guard every denominator with a *pure underflow floor* den = max(den, `eps`(eltype)) (machine epsilon, *not* a problem scale), and do not declare convergence at $k = 0$ — require at least one completed iteration. If $\boldsymbol{u} \equiv 0$ genuinely solves it (zero boundary conditions, $\boldsymbol{f} = 0$), the numerators are also ≈ 0 and the floor yields convergence without inventing a scale.

2. **BC-driven flow, $\boldsymbol{f} = 0$.** The denominator must not rely on $\boldsymbol{f}$; once nonzero the internal terms carry D_M. $\|\boldsymbol{f}\| = 0$ contributing nothing is correct.

3. **Uniform porosity $\alpha \equiv$ const (incl. $\alpha \equiv 1$).** $\nabla\alpha = 0 \Rightarrow \|\nabla(\alpha\boldsymbol{u})\| = \alpha\,\|\nabla\boldsymbol{u}\|$, so $\varepsilon_\mathrm{C} \to \|\nabla\cdot\boldsymbol{u}\| \,/\, \|\nabla\boldsymbol{u}\|$. Expected; do not special-case.

4. **Locally near-uniform flux.** Never a pointwise ratio — always global L^2.

5. $\varepsilon_\mathrm{C} > \sqrt{d}$**.** Analytically impossible (proposition 1) $\Rightarrow$ a bug indicator (assert / log).

6. **All-Dirichlet / pressure indeterminacy ($\Gamma_\mathrm{N} = \varnothing$, $\varepsilon = 0$).** Pressure is defined up to a constant (fixed by the zero-mean condition). The flux-gradient ratio is immune — it uses no $\|p\|$. Do *not* build a pressure-normalized mass measure here.

7. **Compressibility $\varepsilon > 0$ (iterative penalty).** The strict numerator is $\|\varepsilon p^k + \nabla\cdot(\alpha\boldsymbol{u}^k)\|$, but $\varepsilon^* \sim 10^{-4}$, so $\|\nabla\cdot(\alpha\boldsymbol{u}^k)\|$ is equivalent and preferred. The penalty's right-hand-side term cancels at the fixed point.

8. **Newton vs Picard.** Always measure the *genuine nonlinear residual* evaluated at U^k (re-assemble the operator), never the linearized / inner linear-solve residual — the latter $\to 0$ *inside* each step and does not reflect outer convergence. (Gating on the inner residual makes the outer loop stop too early or behave erratically.)

9. **Tolerance vs floor.** ε_{M} and ε_{C} cannot fall below the level set by the discretization (the stabilized scheme permits a small nonzero $\nabla\cdot(\alpha\boldsymbol{u}_h)$, $\mathcal{O}(h^k)$) and by ε. Set `tol` *above* this floor. Diagnostic: a fixed outer-iteration count independent of `tol` signals either `tol` below the floor, or measuring a quantity that does not decrease (edge case 8) — check both.

10. **Stabilization terms.** Numerator = the method's full residual (ASGS/OSGS, subscales included). Denominator = the Galerkin physical force terms of section 3; adding the stabilization terms to D_{M} is optional and harmless (bounded by the Galerkin terms by design).

11. **Element-wise vs global accumulation.** Accumulate $\|\cdot\|^2$ per element, then sum and take the square root (one assembly pass). The global ratio is the default. If a regime varies *strongly* across the mesh, an element-wise envelope (ratio per element, reduced by max or ℓ^2-over-elements) is the most faithful variant; offer only if needed.

6 Optional fallback (documented, NOT the default): pressure-normalized mass measure, still scale-free

If a pressure-referenced mass measure is explicitly wanted (to read the criterion directly as a relative pressure error), replace the global Re, Da by *element-level*

$$Re_h = \frac{|\boldsymbol{u}^k|_K\, h_K}{\nu}, \qquad Da_h = \frac{\sigma_K\, h_K^2}{\alpha_K\, \nu}, \tag{8}$$

the same quantities already inside τ_1 (paper, `eq:Tau1Final`, `eq:TauNavierStokes`). Then

$$\varepsilon_{\mathrm{C}}^{\mathrm{alt}} = \frac{\left\| \left(h^2/(\alpha_K \tau_{1,K})\right) \nabla\cdot(\alpha\boldsymbol{u}^k) \right\|}{\|p^k\|} \sim \frac{\left\| \left(\nu(1+Re_h+Da_h)/\alpha_K\right) \nabla\cdot(\alpha\boldsymbol{u}^k) \right\|}{\|p^k\|}, \tag{9}$$

i.e. the divergence→pressure conversion factor expressed through the solver's own stabilization parameter τ_1 (so still no bespoke scales — only h_K, the iterate, and material data). It is mesh-coupled and fussier than the flux-gradient ratio of section 4 and reduces to it in spirit. Use only if the explicit pressure reading is required; otherwise prefer section 4. **This fallback is intentionally *not* implemented in the code.**

7 Implementation notes

- Reuse the existing residual weak form for r_{M}^k (Philosophy A) so the stabilized residual and the convergence measure stay consistent by construction.
- $L^2(\Omega)$ norm of a field w: $\left(\sum_K \int_K w\cdot w\right)^{1/2}$ (Frobenius inner product for tensors).
- $\nabla\cdot(\alpha\boldsymbol{u})$ and $\nabla(\alpha\boldsymbol{u})$ via the differential operators on the α-field interpolated in the *same* finite element space as the solution (consistent with the paper's nodal interpolation of α).
- Guard denominators with the pure underflow floor den = max(den, `eps`).

- Log ε_{M}, ε_{C}, and each D_{M} term per iteration; the per-term breakdown tells which balance is limiting convergence when the loop stalls.
- Assert $\varepsilon_{\mathrm{C}} \le \sqrt{d}\,(1 + \texttt{tol})$ as a cheap correctness check.

8 Implementation status

The *momentum* (Philosophy A) specification of section 3 is the one used in production, unchanged. The *mass* measure has evolved from the flux-gradient ratio of section 4. Concretely:

- **Gate (production).** The mass gate is the "Route-B Philosophy-A" *algebraic* measure

$$\varepsilon_{\mathrm{C}} = \frac{\left\| r_{\mathrm{C}}^{k} \right\|}{D_{\mathrm{C}}^{k}}, \tag{10}$$

  the norm of the assembled *stabilized* mass residual over a term-magnitude envelope D_{C} — *symmetric with* ε_{M} of eq. (1) and gated at the same $\sim 10^{-9}$ level.

- **Residual-floor accept.** A scale-free `residual_floor_reached` accept is also applied: it accepts machine-floor-converged cells whose ε_{C} cannot reach `tol` because D_{C} collapses for near-divergence-free flow (the mass envelope $\to 0$ when the flow is essentially incompressible).

- **Diagnostic.** The flux-gradient ratio $\varepsilon_{\mathrm{C}} = \left\| \nabla\cdot(\alpha \boldsymbol{u}^{k}) \right\| / \left\| \nabla(\alpha \boldsymbol{u}^{k}) \right\|$ of section 4 (and its $-g$-subtracted variant) is now the *diagnostic* `eps_C_strong`, no longer the gate.

The gate is implemented in `src/solvers/convergence_criterion.jl`.